Jixian Li, Ph.D.

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Biography

As a research associate in the Scalable Computational Intelligence team at TACC, I collaborate with scientists to design and build intelligent systems that improve and automate scientific data analysis workflows. My goal is to make scientific discovery simple and efficient.
Previously, as a postdoctoral research associate at the Scientific Computing and Imaging (SCI) Institute under Dr. Chris Johnson, my work focused on developing innovative visualization methods for scientific data, including computational fluid dynamics, medical imaging, and biomedical simulations.
I received my Ph.D. from the University of Arizona, advised by Dr. Joshua Levine, where my dissertation studied deep learning approaches for exploring collections of visual features of scalar fields.
My research interests include visualization, uncertainty analysis, machine learning, and computational topology related to scientific data analysis. My broader goal is to help scientists to compute, analyze, and communicate their findings.

Education

2022	**Ph.D. Computer Science**	University of Arizona
2017	**M.S. Computer Science**	University of Arizona
2015	**B.S. Computer Science & Mathematics**	University of Arizona

Employment History

2026-· · ·	**Research Associate**	Texas Advanced Computing Center
2022-2026	**Postdoctoral Research Associate**	SCI Institute, University of Utah
2019	**Summer Intern**	Lawrence Livermore National Laboratory
2017-2022	**Research Assistant**	Dept. Computer Science, University of Arizona
2015-2017	**Teaching Assistant**	Dept. Computer Science, University of Arizona

Publications

Forthcoming

1. Kalina Borkiewicz, **Jixian Li**, Joshua A. Levine, and Katherine E. Isaacs, "May (A)I beautify your visualization? expert judgments of acceptable aesthetic alterations," in *2026 IEEE Visualization and Visual Analytics (VIS)*, 2026. arXiv: 2607.00239.
2. Tanner Finken, **Jixian Li**, Bei Wang, Hanqi Guo, and Joshua A. Levine, "Topology-preserving meshing of implicit scalar fields via monotonicity constraints," in *2026 IEEE Visualization and Visual Analytics (VIS)*, 2026. arXiv: 2608.12142.
3. Francesca Samsel, Paola Arriaga, Iris Song, Duncan Young, and **Jixian Li**, *Antarctica: Looking through the ice into the past*, Art of HPC, Super Computing 2026, 2026.

4. Xinyuan Yan, **Jixian Li**, and Bei Wang, "LoCoMapper: Local structural uncertainty analysis of Mapper graphs under cover perturbations," in *Uncertainty Visualization Workshop 2026*, 2026.

Published Journal Articles

1. Joseph Hart, Bart van Bloemen Waanders, **Jixian Li**, Timbwaoga A. J. Ouermi, and Chris R. Johnson, "Hyperdifferential sensitivity analysis with respect to model discrepancy: Prior distributions," *International Journal for Uncertainty Quantification*, vol. 16, no. 1, pp. 51–77, 2026, ISSN: 2152-5080. DOI: 10.1615/Int.J.UncertaintyQuantification.2025059589.
2. **Jixian Li**, Timbwaoga A. J. Ouermi, Mengjiao Han, and Chris R. Johnson, "WebUVisBox: A scenario-based framework for collaborative and accessible 3-D uncertainty visualization on the web," *IEEE Computer Graphics and Applications*, vol. 46, no. 5, pp. 19–31, Sep. 2026, ISSN: 1558-1756. DOI: 10.1109/MCG.2026.3714725.
3. Qiuming Wang, **Jixian Li**, Philip J. Moos, and Alessandro Venosa, "A single-cell atlas and Shiny-based framework for murine lung injury and remodeling," *Frontiers in Bioinformatics*, vol. 6, 2026, ISSN: 2673-7647. DOI: 10.3389/fbinf.2026.1886929.
4. Matthew Berger, **Jixian Li**, and Joshua A. Levine, "A generative model for volume rendering," *IEEE Transactions on Visualization and Computer Graphics*, vol. 25, no. 4, pp. 1636–1650, 2019. DOI: 10.1109/TVCG.2018.2816059.

Conference Proceedings

1. **Jixian Li**, Timbwaoga A. J. Ouermi, Mengjiao Han, and Chris R. Johnson, "Uncertainty tube visualization of particle trajectories," in *2025 IEEE Workshop on Uncertainty Visualization: Unraveling Relationships of Uncertainty, AI, and Decision-Making*, 2025, pp. 1–11. DOI: 10.1109/UncertaintyVisualization68947.2025.00005.
2. Mengjiao Han, Tushar M. Athawale, **Jixian Li**, and Chris R. Johnson, "Accelerated depth computation for surface boxplots with deep learning," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 38–42. DOI: 10.1109/UncertaintyVisualization63963.2024.00009.
3. Mengjiao Han, **Jixian Li**, Sudhanshu Sane, Shubham Gupta, Bei Wang, Steve Petruzza, and Chris R. Johnson, "Interactive visualization of time-varying flow fields using particle tracing neural networks," in *2024 IEEE 17th Pacific Visualization Conference (PacificVis)*, 2024, pp. 52–61. DOI: 10.1109/PacificVis60374.2024.00015.
4. Sangwon Jeong, **Jixian Li**, Chris R. Johnson, Shusen Liu, and Matthew Berger, "Text-based transfer function design for semantic volume rendering," in *2024 IEEE Visualization and Visual Analytics (VIS)*, 2024, pp. 196–200. DOI: 10.1109/VIS55277.2024.00047.
5. **Jixian Li**, Timbwaoga A. J. Ouermi, and Chris R. Johnson, "Visualizing uncertainties in ensemble wildfire forecast simulations," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 84–88. DOI: 10.1109/UncertaintyVisualization63963.2024.00016.

6. Timbwaoga A. J. Ouermi, **Jixian Li**, Tushar M. Athawale, and Chris R. Johnson, "Estimation and visualization of isosurface uncertainty from linear and high-order interpolation methods," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 51–61. DOI: 10.1109/UncertaintyVisualization63963.2024.00012.

7. Timbwaoga A. J. Ouermi, **Jixian Li**, Zachary Morrow, Bart van Bloemen Waanders, and Chris R. Johnson, "Glyph-based uncertainty visualization and analysis of time-varying vector fields," in *2024 IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks*, 2024, pp. 73–77. DOI: 10.1109/UncertaintyVisualization63963.2024.00014.

8. **Jixian Li**, Danielle Van Boxel, and Joshua A. Levine, "Autoencoder-aided visualization of collections of Morse complexes," in *2022 Topological Data Analysis and Visualization (TopoInVis)*, 2022, pp. 18–28. DOI: 10.1109/TopoInVis57755.2022.00009.

9. Zhe Wang, Dylan Cashman, Mingwei Li, **Jixian Li**, Matthew Berger, Joshua A. Levine, Remco Chang, and Carlos Scheidegger, "NeuralCubes: Deep representations for visual data exploration," in *2021 IEEE International Conference on Big Data (Big Data)*, 2021, pp. 550–561. DOI: 10.1109/BigData52589.2021.9671390.

Thesis

1. **Jixian Li**, "Deep learning approaches for exploring collections of visual features of scalar fields," Ph.D. dissertation, The University of Arizona, 2022.

Presentations & Posters

Public Presentations

1. **Jixian Li**, *From coin flips to deep ensembles: A practical guide to machine learning uncertainty*, Invited talk at the AIX@TACC, University of Texas at Austin, 2026.

2. Knut Olsen, Alex Wohlberg, Sarayu Gadi, and **Jixian Li**, *Penetrating the confusion limit of Rubin LSST*, Project milestone presentation at University of Arizona, 2026.

3. **Jixian Li**, Timbwaoga A. J. Ouermi, and Chris R. Johnson, *Visualizing uncertainties in ensemble wildfire forecast simulations*, Oral presentation at the IEEE Workshop on Uncertainty Visualization: Applications, Techniques, Software, and Decision Frameworks, 2024.

4. **Jixian Li**, *Visualizing collections of data*, Invited talk at the University of Utah, 2023.

5. **Jixian Li**, *Deep learning approaches for exploring collections of visual features of scalar fields*, Invited talk at the SCI Institute, University of Utah, 2022.

6. **Jixian Li**, Danielle Van Boxel, and Joshua A. Levine, *Autoencoder-aided visualization of collections of Morse complexes*, Oral presentation at the IEEE Workshop on Topological Data Analysis and Visualization (TopoInVis), 2022.

Posters

1. Joseph Hart, Bart van Bloemen Waanders, **Jixian Li**, Timbwaoga A. J. Ouermi, and Chris R. Johnson, *ASCR AIVIS: Visualization insight for hyper-differential sensitivity analysis*, Poster presented at SCI X (Also presented in ASCR PI meeting in Texas), 2025.

2. Mengjiao Han, **Jixian Li**, Sudhanshu Sane, Shubham Gupta, Bei Wang, Steve Petruzza, and Chris R. Johnson, *Interactive visualization of time-varying flow fields using particle tracing neural networks*, Poster presented at SCI30, 2024.
3. **Jixian Li**, Timbwaoga A. J. Ouermi, and Chris R. Johnson, *Visualizing uncertainties in ensemble wildfire simulations*, Poster presented at SCI30, 2024.
4. Timbwaoga A. J. Ouermi, **Jixian Li**, Tushar M. Athawale, and Chris R. Johnson, *Approximation and visualization of surface uncertainty*, Poster presented at SCI30, 2024.
5. **Jixian Li**, Peer-Timo Bremer, Matthew Larsen, Cyrus Harrison, and Joshua A. Levine, *Integration between BabelFlow and Ascent for in situ data analysis*, Poster presented at the LLNL student poster symposium, 2019.